-: Gulam Mustafa Mid Term Preparation Videos Notes By "Alishba Adnan"

Order and Degree :-

$$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 4y = e^x$$

Double Derivative / order

power / Degree

$\boxed{\begin{array}{l}\text{having order} \to 2 \\ \text{having degree} \to 3\end{array}}$

$$\frac{dy}{dx} + xy + \frac{d^2y}{dx^2} = x$$

Order → 2
degree → 1

$$\frac{d^3x}{dy^3} + xy\frac{dy}{dx} - 4y = e^y$$

Order → 3
Degree → 1

Initial Value Problem (IVP)

$P(t_0) = P_0$ / $y(0) = 5$

Boundary Value Problem (BVP)

$y(0) = 1$, $y'(2P) = -1$

(agar ~~double~~ Derivative ajae tou BVP Ban Jata)

Linear Order of Nth order

$$a_n(t)y_n + a_{n-1}(t)y_{n-1} + \cdots + a_0(t)y = h(t)$$

not e^x, $\ln(x)$ and $\sin(x)$, $\cos(x)$, $\tan(x)$,

Non-Linear

$$x^3 (y'')^3 + x^2 y (y'')^2 + 3xy' + 5y = e^x$$

is taraf e^x, $\ln(x)$, $\cos(x)$, $\sin(x)$, $\tan(x)$

Explicit Equation:

Easily Explain
Simple Equation $= 0$

Separable Equation:

given $\dfrac{dy}{dx} = f(x, y)$

In form:-

$$\frac{dy}{dx} = h(x)\, g(y)$$

$$\frac{dy}{g(y)} = h(x)\, dx$$

Homogenous Equation:

Consider $\dfrac{dy}{dx} = f(x, y)$

If $f(tx, ty) = t^n f(x, y)$ where t is any number and function is said to be homogenous.

Trick:-

$$\frac{x^1 y^1}{x^2 + y^2}$$

Degree
Multiply → add
addition → no add

So,
$x^1 y^1$ → Degree $(1 + 1)$ → add → 2
$x^2 + y^2$ → Degree $(2) + (2)$ → no add → 2

Homogenous

Exact Differential Equation:

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

then

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \text{OR} \quad M_y = N_x$$

Where

$$f(x,y) = \frac{-M(x,y)}{N(x,y)}$$

$$M(x,y) = \frac{\partial F}{\partial x} \quad \text{and} \quad N(x,y) = \frac{\partial F}{\partial y}$$

formula:-

$$\int M\,dx + \int (\text{term free from } x)\,dy = C$$

$\boxed{\text{Note}}$

Explicit Equation $\rightarrow$ $F(x,y) = 0$ — also homogenous

Implicit Equation $\rightarrow$ $F(x,y) = C$

["0" ke ilawa koi bh variable]

Integerating Factor Technique:

If $M(x,y)\,dx + N(x,y)\,dy = 0$

then $M_y \neq N_x$ (It is not exact then we have to find integerating Factor)

Case 1:

$$I.F = e^{\int \left(\frac{M_y - N_x}{N}\right) dx} = \text{term of } x \,/\, 0$$

Case 2:

$$I.F = e^{\int \left(\frac{N_x - M_y}{M}\right) dy} = \text{term of } y \,/\, 0$$

Case III $\quad I.F = \dfrac{1}{xM+yN}$ where $xM+yN \neq 0$

(It is homogenous)

Case IV

$I.F = \dfrac{1}{xM-yN}$ where $xM-yN \neq 0$

formula:-

I.F (Not Exact Equation) = I.F × any Factor

then said to Exact

then $\quad M_y = N_x$

again apply Formula

$$\int M\,dx + \int (\text{term from from } x)\,dy = C$$

Differential Equation of 1st Order / First Order Linear Equation

$$\frac{dy}{dx} + p(x)\,y = q(x)$$

linear in "y"

center ma bh term of x

Example: $\quad \dfrac{dy}{dx} + (x^2+x)\,y = x^3 + e^x$

terms of x

It is linear in y

single variable in y

$I.F = e^{\int y\,dy}$

$I.F = e^{\frac{y^2}{2}}$

(single variable y ka I.F nikalna)

Bernoulli Equations:

General form : $\dfrac{dy}{dx} + P(x)\,y = q(x)\,y^n$

sirf yeah change

Example: $\frac{dy}{dx} + \frac{1}{x} y = x y^2$

where

$P(x) = \frac{1}{x}$

$q(x) = x$

$y^n = y^2$

(single variable) $y = y$

Substitutions: $u = \cancel{y}/x$

$y = ux$

$\frac{dy}{dx} = v + x \frac{dv}{dx}$

Orthogonal Trajectories 1st Order:

$f(x, y, c) = 0$

nth order

$F(x, y, c, c_1, \ldots, c_n) = 0$

If given:-

$\frac{dy}{dx} = f(x, y)$

Family of Orthogonal

$\frac{dy}{dx} = -\frac{1}{f(x,y)}$

Lec# 11

Radioactive Decay:

$A \longrightarrow$ Amount of radioactive substance

$t \to$ time

Rate of change $\to A(t)$
with respect to time

(Directly proportional)

$$\frac{dA}{dt} = KA$$

where K is a constant proportionality

Population Growth Model:

Initial Value

$$A(t=0) = A(0) = A_0$$

then Solution is

$$A(t) = A_0 e^{Kt} \quad \text{—} (1)$$

Half-life matrial $\to$ K

and also

Half life function:

$$A(T) = \frac{A_0}{2} \quad \text{—} (2)$$

By Comparing (1) and (2)

$$\cancel{A_0} e^{Kt} = \frac{\cancel{A_0}}{2}$$

$$e^{Kt} = \frac{1}{2} = 2^{-1}$$

$$\ln e^{Kt} = \ln 2^{-1}$$

$$Kt \ln e = -1 \ln 2$$

$$Kt (1) = \ln 2$$

$$Kt = \ln 2$$

Quiz C-14 → Half Life → 5568 ± 30 years

Example:

Half life of Isotopes = 16 years
Weight for 30 days = 30 years

we have to find
Initial Value :-

we know that

$$\boxed{\frac{dA}{dt} = A(t) = 30}$$

$$\frac{dA}{dt} = KA \quad \text{with} \quad A(t) = 30$$

find $A_0 = ?$

formula:-

$$A(t) = A_0 e^{kt}$$

putting Values

$$\boxed{\frac{A_0}{2} = KA}$$

$$30 = A_0 e^{kt} \longrightarrow (1)$$

$$Kt = -\ln 2 \quad (\text{Default})$$

$$K(16) = -\ln 2$$

$$K = -\frac{\ln 2}{16}$$

By using $A(t) = 30$
put in (1)

So;

$$30 = A_0 e^{-\frac{\ln 2}{16} \times 30} \Rightarrow 30 = A_0 e^{-\frac{30 \ln 2}{16}}$$

$$\frac{30}{e^{-\frac{30 \ln 2}{16}}} = A_0$$

$$A_0 = 110.04 \text{ g}$$

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Example#3 Pg#83

$T_m = 60°F$ and $T(t_1) = 80°F$

$T(t_0) = 75°F$

Temperature of dead Body was not sick

$T(0) = 98.6°F = T_0$

Solution:-

$$\frac{dT}{dt} = K(T - T_m)$$

$$= K(T-60)$$

and

$$T(0) = T_0 = 98.6°$$

we know that formula:-

$$\frac{T(t_1) - T_m}{T(t_0) - T_m} = e^{K(t_1 - t_0)} \quad (1)$$

putting Values;

But $t_1 - t_0 = 2$ hours:

$$\Rightarrow \frac{80-60}{75-60} = e^{K(2)}$$

$$\Rightarrow \frac{20}{15} = e^{2K} \Rightarrow \frac{4}{3} = e^{2K}$$

Take log on Both side;

$$\ln\left(\frac{4}{3}\right) = \ln e^{2K}$$

$$\ln\left(\frac{4}{3}\right) = 2K \cancel{\ln e}$$

$$\ln\left(\frac{4}{3}\right) = 2K$$

$$\frac{\ln\left(\frac{4}{3}\right)}{2} = K$$

$K = 0.1438$

Now we have to find value of $(t_1 - t_0)$ from eq (1) again putting all values in eq(1)

$$\frac{T(t_1) - T_m}{T(t_0) - T_m} = e^{(t_1 - t_0)}$$

Now we put $T(t_1) = 98.6°F$, So:-

$$\frac{98.6 - 60}{80 - 60} = e^{Kt}$$

$$\ln\left(\frac{38.6}{20}\right) = \ln e^{Kt_2}$$

$$\ln\left(\frac{38.6}{20}\right) = Kt_2 \ln e$$

$$\frac{\ln\left(\frac{38.6}{20}\right)}{K} = t^2$$

$$t_2 = \frac{\ln\left(\frac{38.6}{20}\right)}{0.1438}$$

$t_2 = 4.5730 \approx$ 5 hours and 30 minutes

then

Time of death

$= (12 + 2 + 5)h + 30$ min

$=$ 7:30 approximately.

Lec#12

Application of Non-Linear Equation:

Function:

$$P(t) = P_0 e^{Kt}$$

$\Downarrow$ initial value of population

Case 1:

If $K > 0$ then;

$$\lim_{t \to \infty} P(t) = +\infty \quad \text{(answer } +\infty\text{)}$$

Case 2:

If $K < 0$

then $\lim_{t \to 0} P(t) =$ any no.

Logistic Equation: (Log wali equations)

$$\frac{1}{P}\frac{dP}{dt} = a - bP \quad \text{or} \quad \frac{dP}{dt} = P(a - bP)$$

where b is constant of proportionality.

$-bP^2, \; b > 0 \longrightarrow$ inhibition term

(also called Verhulst-Pearl model)

For Singular Solution:

put $P(a - bP) = 0$

$P = 0$	$a - bP = 0$
	$a = bP$
	$bP = a$
	$P = \frac{a}{b}$

Special cases of logistic Equations:

① Epidemic Spread $\rightarrow \frac{dx}{dt} = kxy$

② Modification of L.E $\rightarrow \frac{dP}{dt} = P(a - b\ln P)$

Chemical Reactions:

$\frac{dX}{dt} = KX$ where $K < 0$ because X is decreasing

M.M.Imp Lec# 13
Short + Quiz + Long.

Higher Order Linear Differential Equations:

General form:-

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_0(x)y = g(x)$$

(Decrease hota Jaega until 1st derivative)

Examples.

① $y'' - 4y = 12x$ with $y(0) = 4$, $y'(0) = 1$
(Linear with IVP)

② $\frac{d^2y}{dx^2} = 12e^{2x} + 4e^{-2x}$

$\boxed{\text{Its Solution should be unique.}}$

Note:-

as we talk about linear so $g(x) = P(x)$, $g(x)$ should be polynomial.

- Not exponential function.
- Not $\ln$
- Not $\cos(x)$, $\sin(x)$, $\tan(x)$, ...

and if we put $g(x)=0$, then it is Higher Order Linear Differential homogenous Equation.

if $g(x)$ = constant = polynomial = 0

if $e^x = \ln x = \sin(x)$ then continous at specific point.

Example;

Given Data.

$$y = cx^2+x+3 \quad \text{①}$$

with initial value problem $x^2y'' - 2xy' + 2y = 6$ ②

$y(0) = 3$, $y'(0) = 1$

Solution:-

Diff w.r.t "x" eq ①

$$\frac{dy}{dx} = 2cx + 1 + 0$$

$$\boxed{\frac{dy}{dx} = 2cx + 1}$$

Again differentiate:-

$$\boxed{\frac{d^2y}{dx^2} = 2c}$$

Now put all values in eq ② answer should be R.H.S = 6.

$$x^2(2c) - 2x(2cx+1) + 2(cx^2+x+3) = 6$$

$$2cx^2 - 4cx^2 - \cancel{2x} + 2cx^2 + \cancel{2x} + 6 = 6$$

$$\cancel{4cx^2} - \cancel{4cx^2} + 6 = 6$$

$$6 = 6 \quad \text{(True)}$$

Now put IVP:-

put IVP in eq ①

(true)

$y(0) = 3 \Rightarrow cx^2 + x + 3 \rightarrow c(0) + 0 + 3 = 3$

$y'(0) = 1 \Rightarrow 2cx + 1 \Rightarrow 2c(0) + 1 = 1$

(true)

So, all satisfied then it is not unique

Imp Difference b/w IVP and BVP

IVP	BVP
$y(a) = b$	$y(a) = y_0$ and $y(b) = y_1$
$y'(a) = b$	$y'(a) = y_0'$ and $y'(b) = y'$
$y''(a) = b$	$y''(a) = y_0''$ and $y''(b) = y_0''$
(1 de Jaege Bs)	(Pairs given)

Now with BVP Example:

Pg#98:–

Given $y = c_1 \cos 4x + c_2 \sin 4x$ — ①

and Boundary Value Problem

$y'' + 16y = 0$ ② , $y(0) = 0$, $y\left(\frac{\pi}{2}\right) = 0$

Diff w.r.t "x" eq ①

$y' = c_1 4(-\sin 4x) + 4c_2(\cos 4x)$

$= -4c_1 \sin 4x + 4c_2 \cos 4x$

$y'' = -4c_1(\cos 4x)(4) + 4c_2(-\sin 4x)(4)$

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$y'' = -16c_1 \cos 4x - 16c_2 \sin 4x$

$y'' = -16(c_1 \cos 4x + c_2 \sin 4x)$

By using eq ① we get;

$y'' = -16y$

$y'' + 16y = 0$ (which is true)

$\boxed{\text{OR:}}$

put all values in eq ②

$-16y + 16y = 0$

$0 = 0$ (true)

Now for BVP:-

we use $y(0) = 0$, $x = 0, y = 0$ in eq ①

$0 = c_1 \cos 4(0) + c_2 \sin 4(0)$

$0 = c_1(1) + c_2(0)$

$0 = c_1 + 0$

$\boxed{0 = c_1}$ put in eq ① (true)

$y = (0) \cos 4x + c_2 \sin 4x$

$y = c_2 \sin 4x$

Now apply $y(\frac{\pi}{2}) = 0$

$x = \frac{\pi}{2}, y = 0$

$0 = c_2 \sin 4\left(\frac{\pi}{2}\right)$

$0 = c_2 \sin 2\pi$

$0 = 0$

Satisfy. But c_2 value does not find, then infinite Solutions.

Linear Dependence:-

$\{f_1(x), f_2(x), \dots f_n(x)\}$

Now multiply with constant not all zero, $C_n \neq 0$

$\{c_1 f_1(x) + c_2 f_2(x) + \dots + C_n f_n(x)\}$

Linear Independence

$$c_1 = c_2 = c_3 = \dots = c_n = 0$$

Most VIP:

<u>Wronskian Method</u>:-

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$= y_1 y_2' - y_1' y_2$$

<u>OR</u>

$$W(f_1(x), f_2(x)) = \begin{vmatrix} f_1(x) & f_2(x) \\ f_1'(x) & f_2'(x) \end{vmatrix}$$

$$= f_1(x) f_2'(x) - f_1'(x) f_2(x)$$

<u>Note</u>:-

- If $W(y_1, y_2) = 0$ Linearly Dependenent
  $W(f_1, f_2, \dots, f_n) = 0$
- But <u>Conversely may or may not be.</u>

  Imp

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Example 3: $y_1 = e^{ax}\cos\beta x$, $y_2 = e^{ax}\sin\beta x$

Linearly Dependent or Independent = ?

$$= \begin{vmatrix} e^{ax}\cos\beta x & e^{ax}\sin\beta x \\ -\beta e^{ax}\sin\beta x + ae^{ax}\cos\beta x & \beta e^{ax}\cos\beta x + ae^{ax}\sin\beta x \end{vmatrix}$$

$y_1 = e^{ax}\cos\beta x$

(product Rule)

$= e^{ax}(-\sin \beta x)(\beta) + (\cos\beta x)(a)(e^{ax})$

$= -\beta e^{ax}\sin\beta x + ae^{ax}\cos\beta x$

$y_2 = e^{ax}\sin\beta x$

(Product Rule)

$= e^{ax}(\cos\beta x)(\beta) + ae^{ax}\sin\beta x$

$$= (e^{ax}\cos\beta x)(\beta e^{ax}\cos\beta x + ae^{ax}\sin\beta x) - e^{ax}\sin\beta x(-\beta e^{ax}\sin\beta x + ae^{ax}\cos\beta x)$$

$$= \beta e^{2ax}\cos^2\beta x + \cancel{ae^{2ax}\cos\beta x\sin\beta x} + \beta e^{2ax}\sin^2\beta x - \cancel{ae^{2ax}\cos\beta\sin\beta x}$$

$$= \beta e^{2ax}(\cos^2\beta x + \sin^2\beta x)$$

$$= \beta e^{2ax}(1)$$

$$= \beta e^{2ax} \neq 0$$

Linearly Independent

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Example $y_1 = x$, $y_2 = x\ln x$
$y_3 = x^2 \ln x$

Wrinskein = ?

$$W(y_1, y_2, y_3) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

3x3 matrix

- $y_1 = x$, $y_1' = 1$, $y_1'' = 0$
- $y_2 = x\ln x$, $y_2' = \cancel{x}\left(\frac{1}{\cancel{x}}\right) + \ln x(1)$

$\boxed{y_2' = 1 + \ln x}$

$y_2'' = 0 + \frac{1}{x} = \boxed{y_2'' = \frac{1}{x}}$

- $y_3 = x^2 \ln x$

$y_3' = x^{\cancel{2}}\left(\frac{1}{\cancel{x}}\right) + \ln \cancel{x}(2x)$

$y_3' = x + \ln x(2x)$

$y_3' = x + 2x\ln x$

$y_3'' = 2\left[(x)(\ln x)\right] + [x]$

$= 2\left[\cancel{x}\left(\frac{1}{\cancel{x}}\right) + \ln x\right][+1] = \cancel{2}\left[2 + \cancel{\ln x}\right]$

$= 2[1 + \ln x) + 1$

$= 2 + 2\ln x + 1$

$= 3 + 2\ln x$

put all value in matrix :-

$$\begin{vmatrix} x & x\ln x & x^2 \ln x \\ 1 & 1+\ln x & 2x\ln x + x \\ 0 & \frac{1}{x} & 2\ln x + 3 \end{vmatrix}$$

Now:-

$$= x \begin{vmatrix} 1+\ln x & 2x\ln x + x \\ \frac{1}{x} & 2\ln x + 3 \end{vmatrix} - x\ln x \begin{vmatrix} 1 & 2x\ln x + x \\ 0 & 2\ln x + 3 \end{vmatrix} + x^2 \ln x \begin{vmatrix} 1 & 1+\ln x \\ 0 & \frac{1}{x} \end{vmatrix}$$

$$= x\left[(1+\ln x)(2\ln x + 3) - \left(\frac{1}{x}\right)(2x\ln x + x)\right] - x\ln x\,[(1)(2\ln x + 3) - 0] + x^2 \ln x\left[\frac{1}{x} - 0\right]$$

$$= x\left[2\ln x + 3 + 2(\ln x)^2 + 3\ln x - \frac{2\cancel{x}\ln x}{\cancel{x}} - \frac{1}{x}\right] - x\ln x\,[2\ln x + 3] + x^{\cancel{2}}\ln x\left(\frac{1}{\cancel{x}}\right)$$

$$= \cancel{2x\ln x} + 3x + \cancel{2x(\ln x)^2} + \cancel{3x\ln x} - \cancel{2x\ln x} - X - \cancel{2x(\ln x)^2} - \cancel{3x\ln x} + x\ln x$$

$= 3x - X + x\ln x \Rightarrow 2x + x\ln x \neq 0$

linearly independent

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lec#14

How to find Solution in General:-

~~Case~~

Linear Independence

$$W(f_1, f_2, \dots f_n) \neq 0$$

Linear Dependence

$$W(f_1, f_2, f_3, \dots f_n) = 0$$

Example $\boxed{Pg\ 12}$

$$y_1 = e^{3x} \quad , \quad y_2 = e^{-3x}$$

$$W(y_1, y_2) \begin{vmatrix} e^{3x} & e^{-3x} \\ 3e^{3x} & -3e^{-3x} \end{vmatrix}$$

$$= e^{3x}(-3e^{-3x}) - e^{-3x}(3e^{3x})$$

$$= -3e^{-3x+3x} - 3e^{-3x+3x}$$

$$= -3e^{0} - 3e^{0}$$

$$= -3 - 3$$

$$= -6 \neq 0 \quad \text{(Independent)}$$

So,

General Solution is

$$y = c_1 e^{3x} + c_2 e^{-3x}$$

$$\boxed{y = c_1 y_1 + c_2 y_2}$$

Complementary function for non-homogenous Linear differential Equations:-

General Solution = Complimentary Solution + any particular Solution.

i.e;

$$y = y_c - y_p$$

Example $y_p = -\frac{11}{12} - \frac{1}{2}x$ and also.

$$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 3x$$

for Complimentary Solution:-

$$D^3y - 6D^2y + 11Dy - 6y = 3x$$

$$y(D^3 - 6D^2 + 11D - 6) = 3x$$

(y and 3x khatam kardyna)

$$D^3 - 6D^2 + 11D - 6 = 0 \longrightarrow (1)$$

Since $D = 1$ is root of eq then:-

we have to $0=0$ for eq (1)

if put $D = 0$	Put $D = 1$
$0 - 6(0) + 11(0) - 6 = 0$	$(1)^3 - 6(1) + 11(1) - 6 = 0$
$0 - 0 + 0 - 6 = 0$	$1 - 6 + 11 - 6 = 0$
$-6 = 0$ $\boxed{\text{Not used}}$	$-11 + 11 = 0$
	$0 = 0$ used

So, $D = 1$ is root

then By using Synthetic Division

Synthetic Division :-

1	1	-6	11	-6
		1	-5	6
	1	-5	6	$\boxed{0}$

Depressed equation:

$$D^2 - 5D + 6 = 0$$
$$D^2 - 3D - 2D + 6 = 0$$
$$D(D-3) - 2(D-3) = 0$$
$$(D-3)(D-2) = 0$$
$$\boxed{D=3} \quad \boxed{D=2}$$

Now $D = 1, 2, 3$

$$\boxed{\begin{aligned} m_1 &= 1 \\ m_2 &= 2 \\ m_3 &= 3 \end{aligned}}$$

Note

① Real and Distinct Roots

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \cdots + c_n e^{m_n x}$$

② Real and Same Roots

$$y_c = (c_1 + c_2 x + c_3 x^2) e^{mx}$$
$$y_c = (c_1 + c_2 x) e^{mx}$$

③ Imaginary and Distinct Roots:

$$y_c = (c_1 \cos \beta x + c_2 \sin \beta x) e^{\alpha x}$$

④ Imaginary and Same Roots:

$$y_c = \{(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x\} e^{\alpha x}$$

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Lec# 15

Working Rule $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = f(x)$

If given:

$$y'' + P(x)y' + Q(x)y = 0$$

Note

If y is given we have to find y_2

Formula:-

$$y_2 = y_1 \int \frac{e^{-\int P\,dx}}{(y_1)^2}(dx)$$

Example:- Pg (131)

$y_1 = x^2$ and equation $x^2y'' - 3xy' + 4y = 0$ — ①

Divide x^2 in eq ①

$$y'' - \frac{3}{x}y' + \frac{4}{x^2}y = 0 \quad \text{—— ②}$$

We know that

$$y'' + P(x)y' + Q(x)y = 0 \quad \text{— ③}$$

Compare ② and ③

$$P = \frac{-3}{x^2}$$

formula:-

$$y_2 = y_1 \int \frac{e^{\int P\,dx}}{(y_1)^2}dx$$

Putting Values;

$$y_2 = x^2 \int \frac{e^{\int -\frac{3}{x^2}dx}}{(x^2)^2}dx$$

$$= x^2 \int \frac{e^{-3\ln x}}{x^4}dx$$

$$= x^2 \int \frac{\cancel{x^3}}{x^{\cancel{4}}}dx$$

$$= x^2 \int \frac{1}{x}dx$$

$$y_2 = x^2 \ln x$$

General Solution:-

$$y = c_1 y_1 + c_2 y_2$$

Put values;

$$y = c_1 x^2 + c_2 x^2 \ln x$$

Quiz:-

$$\int x^2 dx = \frac{x^{2+1}}{2+1} + c$$

$$= \frac{x^3}{3} + c$$

But

$$\int \frac{1}{x}(dx) = \int (x)^{-1} dx = \infty$$

So, we take natural log;

$$= \ln x + c$$

Lec #16

① Example (Pg 136) $2y'' - 5y' - 3y = 0$

$$2D^2y - 5Dy - 3y = 0$$

$$y(2D^2 - 5D - 3) = 0$$

$\boxed{-6}$

$-6+1$

-5

The Indicated Equation:-

$$2D^2 - 5D - 3 = 0$$

$$2D^2 - 6D + D - 3 = 0$$

(factorization)

$$2D(D-3) + 1(D-3) = 0$$

$\boxed{D = 3}$ $\boxed{D = -\frac{1}{2}}$

As D is · Different · Real so, Case 1.

$$y = c_1 e^{3x} + c_2 e^{-\frac{1}{2}x}$$

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Example 2: $y'' - 10y' + 25y = 0$

$$D^2y - 10Dy + 25y = 0$$
$$y(D^2 - 10D + 25) = 0$$

The Indicated Equation:-

$$D^2 - 10D + 25 = 0$$
$$D^2 - 5D - 5D + 25 = 0$$
$$D(D-5) - 5(D-5) = 0$$
$$\boxed{D = 5} \quad \boxed{D = 5}$$

Repeated / Real and Same Roots, So:-

$$y_c = (c_1 + c_2 x) e^{mx}$$

So,

$$y = (c_1 + c_2 x) e^{5x}$$ general Solution

Lec#17

Example 1 (Pg 135)

Find General Solution = ?
$y_c = ?$
$y_p = ?$

$$y'' + 4y' - 2y = 2x^2 - 3x + 6$$

$$D^2y + 4 \cdot Dy - 2y = 2x^2 - 3x + 6$$
$$y(D^2 + 4D - 2) = 2x^2 - 3x + 6$$

Associated Equation is

$$D^2 + 4D - 2 = 0$$ (Factorization Not possible

Here $a = 1$, $b = 4$, $c = -2$ (So apply Quadratic formula)

$$D = \frac{\pm b \pm \sqrt{b^2 - 4ac}}{2a}$$

putting values;

$$= \frac{-4 \pm \sqrt{(4)^2 - 4(1)(-2)}}{2(1)}$$

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$$D = \frac{-4 \pm \sqrt{16+8}}{2} = \frac{-4 \pm \sqrt{24}}{2}$$

$$= \frac{-4 \pm \sqrt{2\times2\times2\times3}}{2} = \frac{-4 \pm 2\sqrt{6}}{2}$$

$$= \frac{\cancel{2}\{-2 \pm \sqrt{6}\}}{\cancel{2}} = -2 \pm \sqrt{6}$$

$$\boxed{D = -2+\sqrt{6}} \qquad \boxed{D = -2-\sqrt{6}}$$

Case 1:-

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

putting values;

$$y = c_1 e^{(2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x}$$

Now find $y_p = ?$

So, $g(x) = 2x^2 - 3x + 6$ (Quadratic)

Assume that

$$y_p = Ax^2 + Bx + C \quad \text{—— (1)}$$

then;

$$y = Ax^2 + Bx + C$$

$$y' = 2Ax + B$$

$$y'' = 2A$$

Now; given Question:-

$$y'' + 4y' - 2y = 2x^2 - 3x + 6$$

putting all values;

$$2A + 4(2Ax + B) - 2(Ax^2 + Bx + C) = 2x^2 - 3x + 6$$

$$2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C = 2x^2 - 3x + 6$$

By Comparing; $\boxed{\text{for } x^2}$

$-2A = 2$

$$\boxed{A = -1}$$

$\boxed{\text{for } x :-}$

$8A - 2B = -3$

$8(-1) - 2B = -3$

$\frac{-8+3}{2} = B$

$$\boxed{B = -\frac{5}{2}}$$

for Constant

$2A + 4B - 2C = 6$

$2(-1) + \cancel{4}^{2}\left(-\frac{5}{\cancel{2}}\right) - 2C = 6$

$-2 - 10 - 2C = 6$

$\frac{-2 - 10 - 6}{2} = C$

$$\boxed{C = -9}$$

~~For General Solution~~:-

$y_p = Ax^2 + Bx + C$

$= (-1)x^2 + \left(-\frac{5}{2}\right)x + (-9)$

$y_p = -x^2 - \frac{5}{2}x - 9$

General Solution $= c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$

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Lec# 19

V.I.P

Example

For y_c

$$\frac{d^3y}{dx^3} - 4\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 5x^2 - 6x + 4x^2e^{2x} + 3e^{5x}$$

(Higher Order Equation)

$$D^3y - 4D^2y + 4Dy = 5x^2 - 6x + 4x^2e^{2x} + 3e^{5x}$$

The associated Equation:

$$D^3 - 4D^2 + 4D = 0$$

$$D(D^2 - 4D + 4) = 0$$

$D = 0$ $\qquad$ $D^2 - 4D + 4 = 0$

$$D^2 - 2D - 2D + 4 = 0$$

$$\boxed{D = 2} \quad \boxed{D = 2}$$

So,

$$\boxed{D=0} \quad \boxed{D=2} \quad \boxed{D=2}$$

$$y_c = c_1e^{0x} + (c_2 + c_3x)e^{2x}$$

(Different 2 ayen D ke sath lagega)

For y_p:-

(Degree zaida karde)

$$g(x) = 5x^2 - 6x - 4x^2e^{2x} + 3e^{5x}$$

(Degree zaida)

then Assume:-

$$D^3(5x^2 - 6) = 0$$

$$(D-2)^3 x^2e^{2x} = 0$$

$$(D-5)x^0e^{5x} = 0$$

then associated

$$D^3 - 4D^2 + 4D = 0$$

$\boxed{D=0}$ then Combine

$\boxed{D=0}$ $\quad D^3(D-2)^3(D-5)(D^3 - 4D^2 + 4D) = 0$

$$\boxed{D=0} \quad \boxed{D=2} \quad \boxed{D=5} \quad \boxed{D=2} \quad \boxed{D=2}$$

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(4)

Final $D = 0, 0, 0, 0, 2, 2, 2, 2, 2, 5$

So $y_p = (c_4 + c_5 x + c_6 x^2 + c_7 x^3) e^{0x} + (c_8 + c_9 x + c_{10} x^2 + c_{11} x^3 + c_{12} x^4) e^{2x} + c_{13} e^{5x}$

Hence

$y = y_c + y_p$

lec#20, 21 (Same Method)

Variation of Parameters Method:

Working Rule:

(1) 1st we have to find y_c with the help of D (annihilator)

(2) use $u_K = \int \frac{W_K}{W} dx$ $\{K = 1, 2, \dots, n\}$

★ 1st we find W_K i.e; $W_1, W_2, \dots W_n$ and W (Wornksin Method)

(3) We have to find y_c with help of

$y_c = u_1 y_1 + u_2 y_2 + \dots + u_n y_n$

(4) then we have to putting values in

$y = y_c + y_p$ (General formula)

Example 1: $4y'' + 36y = \operatorname{cosec} 3x$

Solution:-

(Annihilation Method)

$4D^2 y + 36 y = \operatorname{cosec} 3x$

$y(4D^2 + 36) = \operatorname{cose} 3x$

associated equation:-

$4D^2 + 36 = 0$

$4D^2 = -36$

$D^2 = -\frac{\cancel{36}\,\cancel{18}\,9}{4\ 2\ 1} = -9$

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$D^2 = -9$ $\quad \cancel{\sqrt{(D)^2} = \sqrt{(3)}}$ $\quad \cancel{D = \pm 3}$

$D^2 = \sqrt{-1 \times 9} = \sqrt{-1}\sqrt{9} = \pm i3$

$\boxed{D = 0 \pm i3}$ (Imaginary and Distinct Root)

So;

$y_c = (c_1 \cos \beta x + c_2 \sin \beta x) e^{\alpha x}$

$\boxed{y_c = (c_1 \cos 3x + c_2 \sin 3x) e^{0x} = 1}$

Now we have to check linearly

☆ Independent then Let

$y_1 = \cos 3x \quad , y_2 = \sin 3x$

$$W(y_1, y_2) = \begin{vmatrix} \cos 3x & \sin 3x \\ -3\cos 3x & 3\cos 3x \end{vmatrix}$$

$(\cos 3x)(3\cos 3x) - (\sin 3x)(-3\cos 3x)$

$3\cos^2 3x + 3\sin^2 3x$

$3(\cos^2 3x + \sin^2 3x)$

$3(1)$

$\boxed{W = 3} \neq 0$ linearly Independent

☆ Now take $4y'' + 36y = \operatorname{cosec} 3x$

$y'' + 9y = \frac{1}{4} \operatorname{cosec} 3x$

So, $f(x) = \frac{1}{4} \operatorname{cosec} 3x$

☆ find w_1, w_2 :— Quiz

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

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$$W_1 = \begin{vmatrix} 0 & \sin 3x \\ \frac{1}{4}\operatorname{cosec} 3x & 3\cos 3x \end{vmatrix}, \quad W_2 = \begin{vmatrix} \cos 3x & 0 \\ -3\sin 3x & \frac{1}{4}\operatorname{cosec} 3x \end{vmatrix}$$

$$0 - (\sin 3x)\left(\frac{1}{4}\operatorname{cosec} 3x\right), \quad (\cos 3x)\left(\frac{1}{4}\operatorname{cosec} 3x\right) - 0$$

$$-\frac{1}{4}\cancel{\sin 3x}\left(\frac{1}{\cancel{\sin 3x}}\right), \quad \frac{1}{4}\cos 3x \times \frac{1}{\sin 3x}$$

$$\Rightarrow -\frac{1}{4}, \quad \frac{1}{4}\tan 3x$$

So, $\boxed{y_p = (u_1 y_1 + u_2 y_2)}$ Sara kuch y_p ko find karne ke liye

$$\boxed{u_1 = \int \frac{W_1}{W}(dx)}$$

$$= \int \frac{-\frac{1}{4}}{3}(dx) = \int -\frac{1}{4} \times \frac{1}{3}(dx)$$

$$= -\frac{1}{12}\int 1\, x \Rightarrow -\frac{x}{12} \qquad \boxed{u_1 = -\frac{x}{12}}$$

$$u_2 = \int \frac{W_2}{W}(dx)$$

$$= \int \frac{\frac{\cos 3x}{4\sin 3x}}{3}(dx) \Rightarrow \int \frac{\cos 3x}{4\sin 3x} \times \frac{1}{3}(dx)$$

$$= \frac{1}{12}\int \frac{\cos 3x}{\sin 3x}(dx)$$

Minus on all, − and (3) Multiply + divide

$$= \frac{-1}{12} \times \frac{1}{3}\int \frac{-3\cos 3x}{\sin 3x} \Rightarrow -\frac{1}{36}\int \frac{-3\cos 3x}{\sin 3x}$$

$\Rightarrow \frac{-1}{36}\ln(\sin 3x)$

Now; $y_p = u_1 y_1 + u_2 y_2$

$y_p = \left(\frac{-1}{12}x\right)(\cos 3x) + \left[\frac{-1}{36}\ln(\sin 3x)(\sin 3x)\right]$

$$\boxed{y_p = \frac{-x\cos 3x}{12} - \frac{\sin 3x \ln(\sin 3x)}{36}}$$

* General Solution:

$$y = y_c + y_p$$

$$y = c_1\cos 3x + c_2\sin 3x - \frac{x\cos 3x}{12} - \frac{\sin 3x \ln(\sin 3x)}{36}$$

Lec#21

Example:

$$\frac{d^3y}{dx^3} + \frac{dy}{dx} = \csc x$$

(Same as previous)

$$D^3y + Dy = \csc x$$

$$y(D^3 + D) = \csc x$$

Associated Equation

$$D^3 + D = 0$$

$$D(D^2+1) = 0$$

$\boxed{D = 0}$ $\quad D^2 + 1 = 0$

$$D^2 = -1$$

$$\sqrt{D^2} = \sqrt{-1}$$

$$\boxed{D = 0 \pm i}$$

Imaginary and Distinct So,

$$\boxed{y_c = \left(c_1 \cos x + c_2 \sin x\right) e^{0x}}$$

* Now check linearly Independent :-

$$W(y_1, y_2, y_3) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

Since $y_c = c_1 e^{0x} + c_2 \cos x + c_3 \sin x$

$y_1 = 1$, $y_2 = \cos x$, $y_3 = \sin x$

$$W(y_1, y_2, y_3) = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & (-\sin x) & \cos x \\ 0 & (-\cos x) & -\sin x \end{vmatrix}$$

$$= \left(\sin^2 x + \cos^2 x\right) - \cos x\left(0 - 0\right) + \sin x\left(0 - 0\right)$$

$$= 1 - 0 + 0$$

$\boxed{W = 1}$ $\neq 0$ linearly Independent

Now find W_1, W_2, W_3 :- $f(x) = \tan x$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ f(x) & y_2'' & y_3'' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & f(x) & y_3'' \end{vmatrix}$$

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$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & f(x) \end{vmatrix}$$

Now putting values:-

$$W = \begin{vmatrix} 0 & \cos x & -\sin x \\ 0 & -\sin x & \cos x \\ \tan x & -\cos x & -\sin x \end{vmatrix}$$

$(\sin^2 x + \cos^2 x) - 0 + (-\sin x)[\sin^2 x + \cos^2 x]$

$1 \quad - \sin x (1)$

$\Rightarrow 0(\sin^2 x + \cos^2 x) - \cos x (0 - \cos x \tan x) + (-\sin x)(\sin^2 x + \cos^2 x)$

$\Rightarrow 0 - \cos x (-\cos x \tan x) + (\sin x)(1)$

$\Rightarrow \cos^2 x \tan x - \sin x$

Expanding By R_3

$\Rightarrow \tan x (1)$

$\Rightarrow \tan x$

$$W_2 = \begin{vmatrix} 1 & 0 & \sin x \\ 0 & 0 & \cos x \\ 0 & \tan x & -\sin x \end{vmatrix}$$

Expanding by R_1

$= (1)[0 - \cos x \tan x]$

$= -\cos x \tan x$

$= -\cancel{\cos x} \dfrac{\sin x}{\cancel{\cos x}} = -\sin x$

$$W_3 = \begin{vmatrix} 1 & \cos x & 0 \\ 0 & (-\sin x) & 0 \\ 0 & (-\cos x) & \tan x \end{vmatrix}$$

Expanding by R_1

$$= [-\sin x \tan x - 0]$$

$$= -\sin \frac{\sin x}{\cos x} \Rightarrow \frac{-\sin^2 x}{\cos x}$$

Now :-

$$u_1 = \int \frac{W_1}{W} dx$$

$$= \int \frac{\tan x}{1} (dx)$$

$$= -\int -\frac{(\sin x)}{\cos x} (dx)$$

$$u_1 = -\ln(\cos x)$$

$$u_2 = \int \frac{W_2}{W} dx$$

$$= \int \frac{-\sin x}{1} (dx)$$

$$= -\int \sin x (dx)$$

$$= -(-\cos x)$$

$$u_2 = \cos x$$

$$u_3 = \int \frac{W_3}{W} (dx)$$

$$= \int \frac{-\sin x \tan x}{1} (dx)$$

$$= -\int \frac{\sin^2 x}{\cos x} (dx)$$

$$\boxed{\sin^2 x = 1 - \cos^2 x}$$

$$= -\int \frac{1 - \cos^2 x}{\cos x} (dx)$$

$$= -\left[\int \frac{1}{\cos x} - \int \frac{\cos^2 x}{\cos x}\right.$$

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$$\Rightarrow -\left[\int \frac{1}{\cos x}(dx) - \int \cos x(dx)\right]$$

$$\Rightarrow -\left[\int \sec x(dx) - \int \cos x(dx)\right]$$

$$= -\left[\ln|\sec x + \tan x| - \sin x\right]$$

$$= \sin x - \ln|\sec x + \tan x|$$

$$y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$$

Putting values:-

$$y_p = -\ln(\cos x)(1) + (\cos x)(\cos x) + (\sin x - \ln|\sec x + \tan x|)(\sin x)$$

$$y_p = -\ln\cos x + \cos^2 x + \sin^2 x - \sin \ln|\sec x + \tan x|$$

General Solution:-

$$y = y_c + y_p$$

$$y = c_1 e^{0x} + (c_2 \cos x + c_3 \sin x)e^{0x} + 1 - \ln\cos x - \sin\ln|\sec x + \tan x|$$

Required Solution.

Short Questions $\boxed{\text{Lec\#22}}$

$$F = ma$$

$$m = 10kg, \quad a = 2ms^{-1}$$

$$F = (10)(2)$$

$$\boxed{F = 20N}$$

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$m = ?$, $F = 20N$, $a = 2ms^{-2}$

$$F = ma$$

$$= (2)(\ )$$

$$(20) = m(2)$$

$$\frac{20}{\cancel{2}} = m$$

$$\boxed{m = 20\ kg}$$

Amplitud:

$$A = \sqrt{(C_1^2 + C_2^2)} \quad OR$$

$$A = \sqrt{(C_1 + C_2)^2}$$

Imp Time Period

$$T = \frac{2\pi}{\omega}$$

$$\boxed{\pi = 3.14}$$

frequency:

Reciprocal of Time period is called frequency:-

$$\boxed{f = \frac{1}{T} = \frac{\omega}{2\pi}}$$

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